Experiment

Simple Linear Regression Using California Housing Dataset (CSV Input)

Aim

To implement **Simple Linear Regression** using one feature (total_rooms) from the California Housing dataset (housing.csv) to predict house prices using the **top 150 records**.

Objectives

- Load dataset from a CSV file
- Select a single feature and target variable
- Build a regression model manually
- Compute **MSE and R^2 manually**
- Visualize regression line

Theory

◆ **Simple Linear Regression**

$$y = b_0 + b_1 x$$

Where:

- x: Independent variable (total_rooms)
- y: Dependent variable (median_house_value)

Regression Coefficients

$$b_1 = \frac{\sum(x - \bar{x})(y - \bar{y})}{\sum(x - \bar{x})^2}$$

$$b_0 = \bar{y} - b_1 x$$

Mean Squared Error (MSE)

$$MSE = \frac{1}{n}\sum(y_i - \hat{y}_i)^2$$

R-squared (R^2)

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

Procedure

1. Load dataset from housing.csv
2. Select:
   - Feature: total_rooms
   - Target: median_house_value
3. Use **first 150 rows**
4. Compute regression coefficients
5. Predict values
6. Compute MSE and R^2
7. Plot regression line

Python Code

```
import numpy as np

import pandas as pd

import matplotlib.pyplot as plt


# -----------------------------

# Load dataset

# -----------------------------

df = pd.read_csv("/content/sample_data/housing.csv")


# -----------------------------

# Select feature and target

# -----------------------------

X = df['total_rooms'].head(150).values.reshape(-1, 1)
```

```
y = df['median_house_value'].head(150).values.reshape(-1, 1)

# -----------------------------
# Visualization (Scatter Plot)
# -----------------------------
plt.scatter(X, y, color="blue", s=20)

# -----------------------------
# Mean values
# -----------------------------
m_x = np.mean(X)
m_y = np.mean(y)

# -----------------------------
# Compute coefficients
# -----------------------------
SS_xy = np.sum((X - m_x) * (y - m_y))
SS_xx = np.sum((X - m_x) * (X - m_x))

b1 = SS_xy / SS_xx
b0 = m_y - b1 * m_x

print("Slope (b1):", b1)
print("Intercept (b0):", b0)

# -----------------------------
# Predictions
# -----------------------------
```

```
y_pred = b0 + b1 * X

# -----------------------------
# Manual MSE
# -----------------------------
n = len(y)
mse = np.sum((y - y_pred) ** 2) / n
print("MSE:", mse)

# -----------------------------
# Manual R^2
# -----------------------------
SS_res = np.sum((y - y_pred) ** 2)
SS_tot = np.sum((y - m_y) ** 2)

r2 = 1 - (SS_res / SS_tot)
print("R^2:", r2)

# -----------------------------
# Plot regression line
# -----------------------------
plt.plot(X, y_pred, color="red")

plt.xlabel("Total Rooms")
plt.ylabel("Median House Value")
plt.title("Simple Linear Regression (150 Samples)")
plt.show()
```

Output

- Scatter plot of data
- Regression line
- Computed values:
    - Slope (b_1)
    - Intercept (b_0)
    - MSE
    - R^2

Slope (b1): 30.03063864613546

Intercept (b0): 144527.3970744946

MSE: 9070344426.642298

R^2: 0.1494586436630212

Result

The linear regression model was successfully implemented using
the total_rooms feature.
Model performance was evaluated using **MSE and R^2**.

- The model establishes a linear relationship between **total rooms and house price**
- Performance depends on how strongly the feature influences the target
- Using only one feature may limit accuracy